0.1 The exact cyclic-cover source and the simultaneous distribution overlap

Statement E.66 (Proved). All E.62 final values survive the five simultaneous distribution tests. *Retain the actual group $N_B \triangleleft B = C_2 * C_3$, its ordered rank-97 free basis, and the 24 words $f_i \in F'$ of E.1. Let C be the actual image of N'_B in the joint E.62 source, with its displayed ordered class-two coordinates. Thus $|C| = 2^{24}$. For $n \in \{2, 3, 4, 6, 12\}$ use the actual cyclic-cover map of D.5, and put*

$$\psi_n(w) = \big(q_F\delta_n(w), c_n(w) \bmod 6\big), \qquad J = \operatorname{im}\left(N'_B \xrightarrow{(\psi_n)_n} (q_F(F) \times \mathbb{Z}/6)^5\right). \tag{E.66.1}$$

Here $q_F(F)$ is the literal order-96 matrix group in E.6; its unique signed representative modulo four has diagonal entries one. The actual common image

$$Z = \operatorname{im}(N'_B \longrightarrow C \times J) \tag{E.66.2}$$

has order 356 241 767 399 424 and index two in $C \times J$. It is exactly the fibre product of the following two nonzero characters:

$$\begin{aligned} \chi_C(c) &= c_4 + c_5 + c_6 + c_9 + c_{10} + c_{11} + c_{13} + c_{15} + c_{17} \pmod 2, \\ \chi_J((h_n, a_n)_n) &= (h_3)_{21}/2 \pmod 2. \end{aligned} \tag{E.66.3}$$

The indices in the first line are zero-based indices in E.62's ordered binary coordinates; its six central coordinates have zero coefficient. The second line uses the lower-left entry of the specified modulo-four representative, which is even.

For every i and every one of the four final E.62 correction values c over f_i, there is a single $u \in N'_B$ whose E.62 value is c and for which $f_i u$ satisfies all five equations

$$q_F\delta_n(f_i u) = q_F(y)^{c_n(f_i u)} q_F(f_i u) \quad (n = 2, 3, 4, 6, 12) \tag{E.66.4}$$

together with

$$c_4 = 2c_2, \qquad c_6 = c_2 + c_3, \qquad c_{12} = 2c_2 + c_3 \pmod 6. \tag{E.66.5}$$

For each specified correction there are exactly three coefficient tuples in this finite target. Their c_3 coordinates are $0, 2, 4$; their c_2 coordinate is the same fixed even value for the given base point. All 96 final E.62 values, including all values over the extra CM coset, are retained.

There is also an exact comparison with the cyclic-three map into $\mathrm{SL}_2(\mathbb{F}_3)$. Its single-coordinate image on N'_B is the whole group. Keeping all three based cyclic coordinates gives an image of order 1536 with derived subgroup of order 512. Keeping their entire six-coset B/F orbit gives an image of order $2^{27} \cdot 3$ with derived subgroup of order 2^{27}. Both abelianizations are C_3, so the actual common correction image of either one with C is their full direct product.

These are exact necessary finite tests on the specified source. They do not assert generation in an additional distribution refinement or a full profinite GT lift, and do not evaluate $\Delta|_{\widehat{\mathrm{GT}}}$ from E.65.

Proof. We describe the source maps before their finite certificates. Put $H_n = \ker(F \to \mathbb{Z}/n)$, with $x \mapsto 1$, $y \mapsto 0$. The map δ_n sends the actual free basis $x^n, y, xyx^{-1}, \ldots, x^{n-1}yx^{-(n-1)}$ to $x, y, 1, \ldots, 1$. Its x-exponent is the original x-exponent divided by n, and c_n is its y-exponent. The former is therefore exactly zero on F'. D.5 compares this basis with the negative Schreier convention and proves the full-GT distribution equation from Enriquez's theorem.

One actual induced source. Let $A = q_F(F) \times \mathbb{Z}/6$. For each n use the section $1, x, \dots, x^{n-1}$ and the right wreath-product coordinates

$$(k; (a_j)_j)(l; (b_j)_j) = (k + l; (a_j b_{j+k})_j). \tag{E.66.6}$$

The source images are

$$x \longmapsto (1; 1, \dots, 1, (q_F(x), 0)), \qquad y \longmapsto (0; (q_F(y), 1), 1, \dots, 1). \tag{E.66.7}$$

For $w \in H_n$, coordinate j is the value of ψ_n on $x^j w x^{-j}$, so coordinate zero is exactly (E.66.1). Every $u \in N_B'$ belongs to every H_n in question.

The signed order-96 group has the actual faithful action on four points from its modulo-four group and on the eight nonzero row vectors of $\mathbb{F}_3^2$ from its modulo-three matrix. Adjoin the actions of $\mathbb{Z}/6$ on two and three points. This gives a faithful 17-point action of A. Right multiplication on the four-point group and the rule $v \mapsto vh$ on row vectors make its permutation multiplication agree with (E.66.6). The five induced actions have total degree $17(2 + 3 + 4 + 6 + 12) = 459$.

The program reads exactly the 97 original N_B words used in E.62 and forms their common block action on these 459 points and E.62's 4764 points. It computes the full derived image of this common source. Only then does it project each shift-zero distribution factor to its coordinate-zero 17-point block. Thus the resulting degree-4849 image is literally Z, rather than a product of independently chosen corrections. Its two projections are the already specified C and J.

Complete Schreier certificates give

$$|J| = 42\,467\,328, \qquad |C| = 16\,777\,216, \qquad |Z| = \frac{|C||J|}{2}. \tag{E.66.8}$$

Every retained generator in the derived certificate is an actual word in commutators and their conjugates from the original N_B basis; all final Schreier relations are checked. This proves both containments required for the displayed image orders.

The common character. Because Z is a subdirect subgroup of index two, define $\chi_C(c)$ to be zero precisely when $(c, 1) \in Z$, and define $\chi_J(j)$ analogously. The two projection-kernel quotients give nonzero homomorphisms to C_2 and $Z = \{(c, j) : \chi_C(c) = \chi_J(j)\}$. Membership of every E.62 ordered basis element gives the first line of (E.66.3). All basis products are independently checked; the character vanishes on the six central coordinates.

For the second line, modulo-four multiplication in the displayed four-element matrix group adds its even lower-left entry after division by two. Hence the proposed formula is a homomorphism on the ambient group. The program decodes every strong generator of J into its five actual A values and compares this formula with $(1, j) \in Z$. They agree on every generator. Finally it checks the two character values on every actual strong generator of Z. This proves the exact formula, including its coordinate convention; it is not inferred solely from the index.

All final fibres and a common word. For each E.62 coordinate c, construct its actual permutation p_c from the frozen ordered class-two basis. For a coefficient tuple $(a_n)_n$, the needed distribution correction is the literal tuple

$$d_n = \psi_n(f_i)^{-1}(q_F(y)^{a_n} q_F(f_i), a_n). \tag{E.66.9}$$

Test $(p_c, (d_n)_n)$ for membership in Z. Enumerating all 36 choices of $(a_2, a_3) \in (\mathbb{Z}/6)^2$ with the other entries defined by (E.66.5) gives exactly three members for each of the 96 specified values. The recorded tuples have $a_3 = 0, 2, 4$ and the fixed even a_2 stated above.

The complete chain also returns a word in its strong generators for every successful membership. Composing its word DAG with the derived-source DAG gives a word in the original 13 selected members of the actual 97-element N_B basis. Each is a product of the already certified N'_B generators. The separate program replays all 288 words in the full degree-5223 common source, not just its coordinate-zero projection, and verifies (E.66.9) and the exact E.62 coordinate. This supplies one common correction for every asserted tuple.

The cyclic-three orbit comparison. Replacing A by the literal $\mathrm{SL}_2(\mathbb{F}_3)$ target with $x \mapsto U^2$, $y \mapsto L^2$ gives

$$x = (1; 1, 1, U^2), \qquad y = (0; L^2, 1, 1). \tag{E.66.10}$$

The 4656 actual pair commutators of the original 97 basis words have coordinate-zero values containing all 23 nonidentity matrices. Thus the single-coordinate image is onto. The full three-coordinate enumeration gives

$$|\operatorname{im} N_B| = 13824, \quad |\operatorname{im} N'_B| = 1536, \quad |\operatorname{im} N''_B| = 512. \tag{E.66.11}$$

The three matrix abelian characters have the diagonal image $\{(a, a, a) : a \in \mathbb{F}_3\}$. Their kernel is exactly the product of the three order-eight matrix kernels, which is the second derived image. This proves the asserted abelianization C_3.

Induce this 72-point action through the six actual cosets of F in B. If t_q is the retained section, the transition at b is the value of $t_q b t_{qb}^{-1}$ in (E.66.10). This gives the actual 432-point B-core. Complete certificates and their independent word replays give

	B	N_B	N'_B	N''_B
|image|	$2^{30}3^7$	$2^{27}3^5$	$2^{27}3$	2^{27}.

(E.66.12)

The last two columns again give abelianization C_3. No identity is imposed on a conjugate coordinate merely from its presence in this induced map.

For each cyclic-three image just obtained, form its joint image with E.62's C using the same domain N'_B. Any common quotient is a two-group. A nontrivial finite two-group has a quotient C_2, whereas all these cyclic-three images have odd abelianization. Their common quotient is therefore trivial, and the actual subdirect image is the direct product. In particular, every specified E.62 correction can be retained while solving the single cyclic-three matrix equation by an additional word in the kernel of its C map.

Certificates and limits of the conclusion. The new files use the prefixes `a_cm_kummer3` and `a_cm_distribution`. Their manifests retain every input, exact source map, word DAG, complete relation check and successful coefficient tuple. The first cyclic-core attempt used all 4600 strong generators as second-derived inputs and was stopped by the memory limit; it is preserved. A complete greedy certificate then retained 16 actual generators of the same group before the successful derived calculation. No incomplete attempt supplies a mathematical conclusion. All old E.59 and E.62 artifacts remain unchanged. The new assertions concern the exact additional finite identities and their common source. They do not replace the remaining full-GT quantifier in E.65. □